

RYAN HERRMANN

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Graduate Mechanical Engineering student with a strong foundation in hands-on engineering and analytical training at CU Boulder. Passionate about designing, building, and validating electromechanical products through prototyping, testing, and simulation. Combining practical engineering experience with data-driven analysis to solve complex technical problems.

EXPERIENCE

Teaching Assistant

CU Boulder

Aug. 2025 – Present

Boulder, CO

- Lead 3 laboratory sections per semester, working directly with 50-70 students per section, for an introductory circuits course
- Taught foundational circuit laws, analysis techniques, and basic circuit design, reinforcing lecture concepts through applied experiments
- Demonstrated and supervised use of laboratory equipment including function generators, multimeters, and oscilloscopes

Field Service Engineer

IQVIA

Apr. 2025 – Aug. 2025

Remote / Nationwide

- Collaborated with Medtronic to upgrade over 50 colonoscopy and endoscopy imaging machines with increased polyp detection and accuracy
- Coordinated software deployment among clinical staff, sales, and internal engineering teams
- Deployed software updates to clinical equipment at 23 facilities nationwide, ensuring uniform functionality across sites

EDUCATION

University of Colorado Boulder

Master of Science, Mechanical Engineering (GPA: 3.88)

Boulder, CO

Aug. 2025 – May 2027

- **Coursework:** Fluid Dynamics, CFD, Design

California Polytechnic State University

Bachelor of Science, Biomedical Engineering, Bioinstrumentation Emphasis (GPA: 3.48)

San Luis Obispo, CA

Sep. 2020 – Jun. 2024

- **Achievements:** President's List (2023-2024): Dean's List honored all 4 quarters
- **Coursework:** Biomechanics, Biomaterials, Biosensors and Circuits, Biopotential Instrumentation, Neural Systems Simulation and Modeling, Fluid Mechanics, Thermodynamics, Linear Analysis II

PROJECTS

Trailhead Safety Indicator | SolidWorks, 3D Printing, Circuit Design, Power Systems

Jan. 2026 – May. 2026

- Collaborated with a team to design and prototype a vehicle-mounted backcountry safety device through a full product development cycle, from user research through prototyping and field testing
- Led the power electronics architecture, achieving 25-day standby life against a 20-day requirement, reducing standby current to ~5 mA
- Designed, built, and refined multiple prototype iterations including a 3D printed enclosure and multiple vehicle mounting systems
- Validated usability through structured user clinics, achieving 80% correct alert-state identification and 70% device visibility at 75 ft

Water Hammer CFD Study | CFD, Python, DOLFINx, FEA

Jan. 2026 – May 2026

- Developed an FSI solver to model water hammer pressure waves in flexible piping systems.
- Validated simulated pressure waves against analytical Joukowsky theory across 9 pipe and fluid combinations
- Co-authored a technical report documenting model performance, limitations, and engineering recommendations.

Biomedical Senior Design | SolidWorks, 3D Printing, Mechanical Design

Sep. 2023 – Mar. 2024

- Designed and prototyped a wrist-worn tremor damping device for individuals with essential tremor or Parkinson's disease, targeting the 5–7 Hz pathological tremor frequency range
- Developed a spring and air damper mechanism housed in a 3D printed enclosure, iterating through 6 design revisions to refine damping performance
- Co-authored a 24-page technical report documenting design requirements, iterative development, and final prototype evaluation

TECHNICAL SKILLS

Programming & Data Analysis: MATLAB, Python, Git, Java, JMP

Design & Simulation: SolidWorks, COMSOL, LTSpice, KiCad, DOLFINx

Tools & Environments: Linux, WSL, tmux, Remote Terminal Workflows